

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0405

CO 3 - ASSESSMENT TOOL 2: Data Interpretation

COURSE OUTCOME COVERED

- Apply data preparation and exploratory data analysis techniques, including data summarization, data distribution analysis, outlier treatment, and measures of asymmetry, to preprocess and interpret datasets effectively.
- Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.

16. Continuous Distribution — Area Interpretation

A continuous random variable X is normally distributed with $\mu = 100$, $\sigma = 15$.

(a) What proportion of values lie between 85 and 130?

(b) A value of 145 is observed — how many standard deviations away is it, and would you flag it as a potential outlier under the empirical rule?

Continuous Distribution — Area Interpretation

Given

- Mean, $\mu = 100$
- Standard deviation, $\sigma = 15$
- X follows a normal distribution.

(a) Proportion of values between 85 and 130

Step 1: Convert values into Z-scores

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

For 85:

$$Z_1 = \frac{85 - 100}{15} = -1$$

For 130:

$$Z_2 = \frac{130 - 100}{15} = 2$$

So we need the area between $Z = -1$ and $Z = 2$.

From the standard normal table:

$$P(Z < 2) = 0.9772$$

$$P(Z < -1) = 0.1587$$

Therefore:

$$P(-1 < Z < 2) = 0.9772 - 0.1587 = 0.8185$$

Answer

81.85%

Approximately 81.9% of the values lie between 85 and 130.

(b) Value 145

Z-score

$$Z = \frac{145 - 100}{15} = 3$$

Therefore, 145 is 3 standard deviations above the mean.

Empirical Rule

For a normal distribution:

- $\pm 1\sigma \rightarrow$ about 68%
- $\pm 2\sigma \rightarrow$ about 95%
- $\pm 3\sigma \rightarrow$ about 99.7%

A value at 3σ is extremely unusual and should be flagged for investigation as a potential outlier, although it is right at the empirical-rule boundary rather than beyond it.

Final Answer

145 is 3 standard deviations above the mean and is a potential extreme/outlier value.

17. Sampling Bias vs Sampling Error

A survey estimating average household expenditure samples only online panel

respondents. The sample mean is ₹18,000 vs a known population mean of ₹14,500 (from census data). (a) Is this discrepancy better explained by sampling error or sampling bias? (b) Would increasing the sample size fix this problem? Justify.

Sampling Bias vs Sampling Error

Given

- Sample mean = ₹18,000
- Population mean = ₹14,500
- Survey includes only online panel respondents

(a) Sampling error or sampling bias?

Definition

Sampling error is the natural difference between a sample result and the population value because only a sample is observed.

Sampling bias occurs when the method of selecting the sample systematically excludes or overrepresents certain groups.

Here, the survey uses only online panel respondents.

People who participate in online panels may not represent the entire population.

Therefore, this is primarily:

Sampling Bias

The difference is:

$$18000 - 14500 = ₹3500$$

(b) Will increasing sample size fix the problem?

Answer

No

Increasing the sample size generally reduces sampling error, but it does not automatically remove sampling bias.

For example, if we survey 100,000 people but all of them come from the same biased online panel, the sample can still systematically differ from the population.

Final Answer

The discrepancy is mainly due to sampling bias. Increasing the sample size alone will not solve it. A more representative sampling method is required.

18. Covariance Sign vs Correlation Strength

Two variable pairs both show positive covariance: Pair 1 has Cov = 850, Pair 2 has Cov = 12. Explain why you cannot conclude Pair 1 has a "stronger relationship" than Pair 2 without additional information, and state what that missing information is.

Covariance Sign vs Correlation Strength

Given

- Pair 1: Covariance = 850
- Pair 2: Covariance = 12

Both covariances are positive.

What does positive covariance mean?

Positive covariance means that, generally, when one variable increases, the other variable also tends to increase.

However, covariance magnitude cannot directly be used to compare relationship strength.

Why?

Because covariance depends on the units and scales of the two variables.

For example:

- Covariance between height and weight
- Covariance between income and expenditure

may have completely different numerical magnitudes because their measurement scales are different.

Therefore:

$850 > 12$

does not mean Pair 1 has a stronger relationship.

What information is missing?

We need the correlation coefficient.

$$r = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Correlation is standardized and ranges from:

$$-1 \leq r \leq 1$$

Final Answer

We cannot conclude that Pair 1 has a stronger relationship because covariance depends on the variables' units and standard deviations.

The missing information is the correlation coefficient, or equivalently the standard deviations of both variables so that correlation can be calculated.

19. Estimation — Confidence Interval Reasoning

A sample of $n = 100$ items has mean = 250 and SD = 40. Construct a 95% confidence interval for the population mean. Then explain, in interpretation (not formula) terms, what this interval does and does NOT tell you about the true population mean.

Estimation — Confidence Interval

Given

- Sample size, $n = 100$
- Sample mean = 250
- Sample SD = 40
- Confidence level = 95%

For a large sample, we can use:

$$CI = \bar{x} \pm 1.96 \frac{s}{\sqrt{n}}$$

Step 1: Standard error

$$SE = \frac{40}{\sqrt{100}}$$
$$SE = \frac{40}{10} = 4$$

Step 2: Margin of error

$$ME = 1.96(4)$$

$$ME = 7.84$$

Step 3: Confidence interval

Lower limit:

$$250 - 7.84 = 242.16$$

Upper limit:

$$250 + 7.84 = 257.84$$

Therefore:

$$95\% CI = (242.16, 257.84)$$

Interpretation

We are 95% confident that the population mean lies between ₹242.16 and ₹257.84.

What it does NOT mean

It does not mean that there is a 95% probability that this particular fixed population mean is inside this already-calculated interval.

Instead, the method used to construct such intervals captures the true population mean about 95% of the time over repeated sampling.

20. Integrated Case Study

A retail company's daily sales data (in ₹'000) for 15 days is:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

(a) Identify the outlier and justify using both the IQR method and z-score method.

(b) Compute mean and median with and without the outlier — compare the sensitivity of each.

(c) Compute skewness with the outlier included — classify the distribution shape.

(d) Recommend whether the company should remove, cap, or investigate the outlier before reporting monthly performance, and justify your recommendation using the statistical evidence from (a)–(c).

Integrated Case Study

Given daily sales in ₹'000

120,125,118,130,122,128,500,119,124,121,126,123,117,129,125

There are 15 observations.

(a) Identify the outlier using IQR

Step 1: Arrange data

117,118,119,120,121,122,123,124,125,125,126,128,129,130,500

Step 2: Find median

The middle value is the 8th value:

$$Q_2 = 124$$

Step 3: Find Q_1

Lower half:

117,118,119,120,121,122,123

Therefore:

$$Q_1 = 120$$

Step 4: Find Q_3

Upper half:

125,126,128,129,130,500

Using the median-of-halves method, with the central overall median excluded, the upper seven are:

125,126,128,129,130,500

Actually, because there are 15 observations, the upper half contains 7 values:

125,126,128,129,130,500

This reveals an important issue: the listed upper half should be the observations 9–15:

125,125,126,128,129,130,500

Thus:

$$Q3=128$$

So:

$$IQR=Q3-Q1$$

$$IQR=128-120=8$$

Step 5: Calculate outlier limits

Lower limit:

$$Q1-1.5(IQR)$$

$$120-1.5(8)=108$$

Upper limit:

$$Q3+1.5(IQR)$$

$$128+1.5(8)=140$$

Therefore:

Acceptable range=108 to 140

Since:

$$500>140$$

500 is an outlier.

Z-score method

Mean including the outlier:

$$\bar{x} = \frac{2227}{15}$$

$$\bar{x} = 148.47$$

Population standard deviation is approximately:

$$\sigma=94.03$$

Z-score of 500:

$$Z=\frac{500-148.47}{94.03}$$

$$Z \approx 3.74$$

Since the absolute z-score is greater than 3:

$$|Z| > 3$$

500 is a clear extreme observation.

Conclusion

Both methods identify:

500 as an outlier

(b) Mean and Median With and Without Outlier

With 500

Mean:

148.47

Median:

124

Without 500

Remaining data:

117,118,119,120,121,122,123,124,125,125,126,128,129,130

Mean:

123.36

Median:

$\frac{2123+124}{2} = 123.5$

Comparison

Measure	With 500	Without 500
Mean	148.47	123.36
Median	124	123.50

The mean changes by approximately:

$$148.47 - 123.36 = 25.11$$

The median changes by only:

$$124 - 123.5 = 0.5$$

Conclusion

The mean is highly sensitive to the outlier, while the median is much more resistant.

(c) Skewness

With the 500 value included, the distribution has a very long tail toward the right.

The calculated skewness is approximately:

3.46

Interpretation

Since:

Skewness > 0

the distribution is:

Highly positively/right skewed

The 500 value creates the long right tail.

(d) Recommendation

The company should investigate the 500 value before removing or capping it.

Why?

Statistical evidence shows:

1. **IQR method:** 500 is above the upper limit of 140.
2. **Z-score:** approximately 3.74, indicating an extreme observation.
3. **Mean:** increases from ₹123.36k to ₹148.47k.
4. **Median:** remains almost unchanged at about ₹123.5k–₹124k.
5. **Skewness:** approximately 3.46, showing strong right skewness.

Recommended action

Do not automatically delete the observation.

First determine whether ₹500k is:

- a genuine unusually high-sales day, or
- a data-entry/measurement error.

If it is a valid business event, retain it and report it separately or use robust measures such as the median.

If it is an error, correct or remove it.

Final recommendation

Investigate first, then decide whether to remove or retain.

Simply deleting a genuine high-sales event could hide important business information.